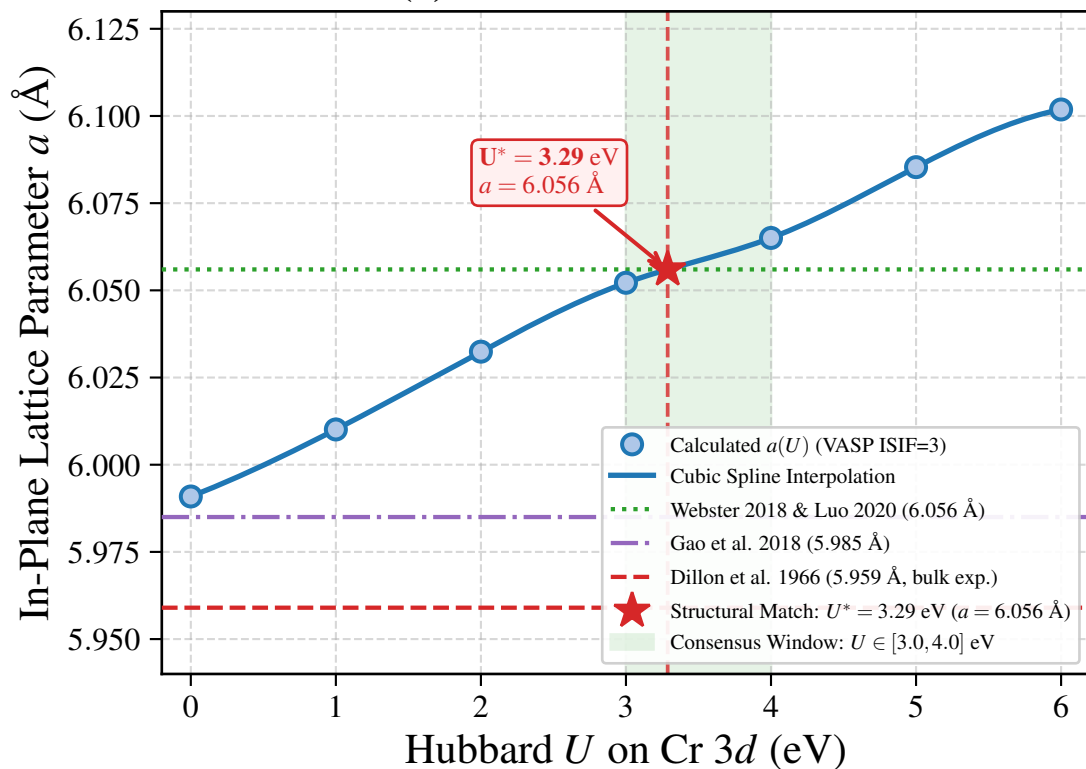
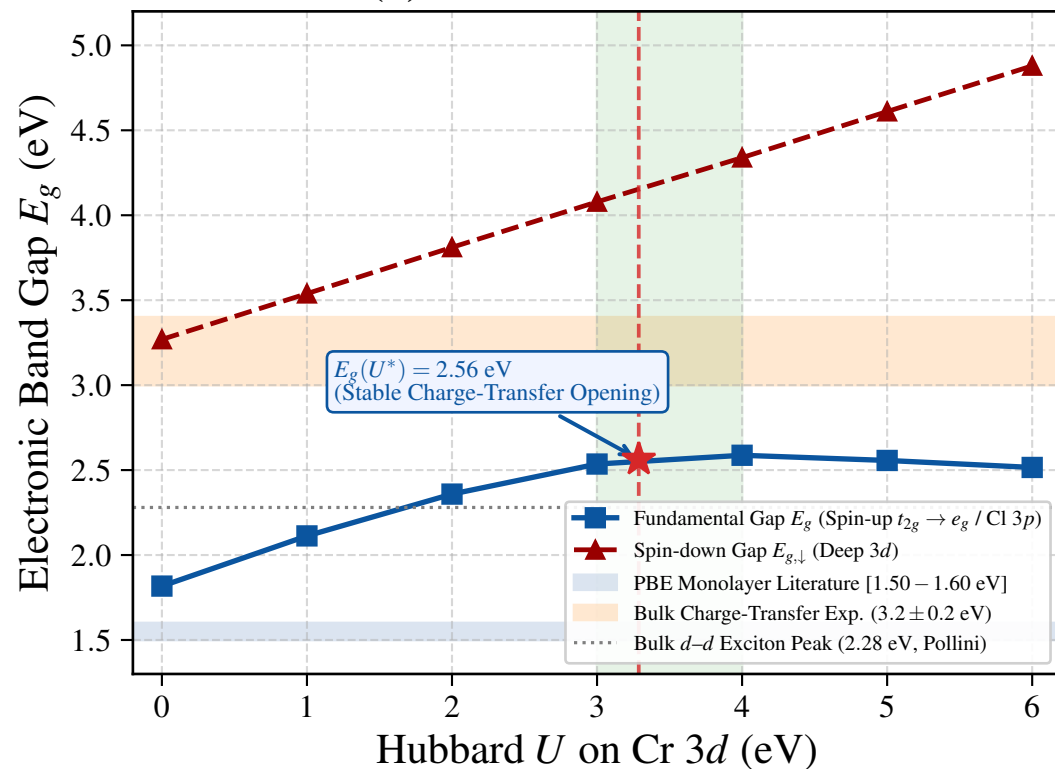


# Monolayer CrCl<sub>3</sub>: Optimal Hubbard $U$ Determination via Multi-Constraint Optimization

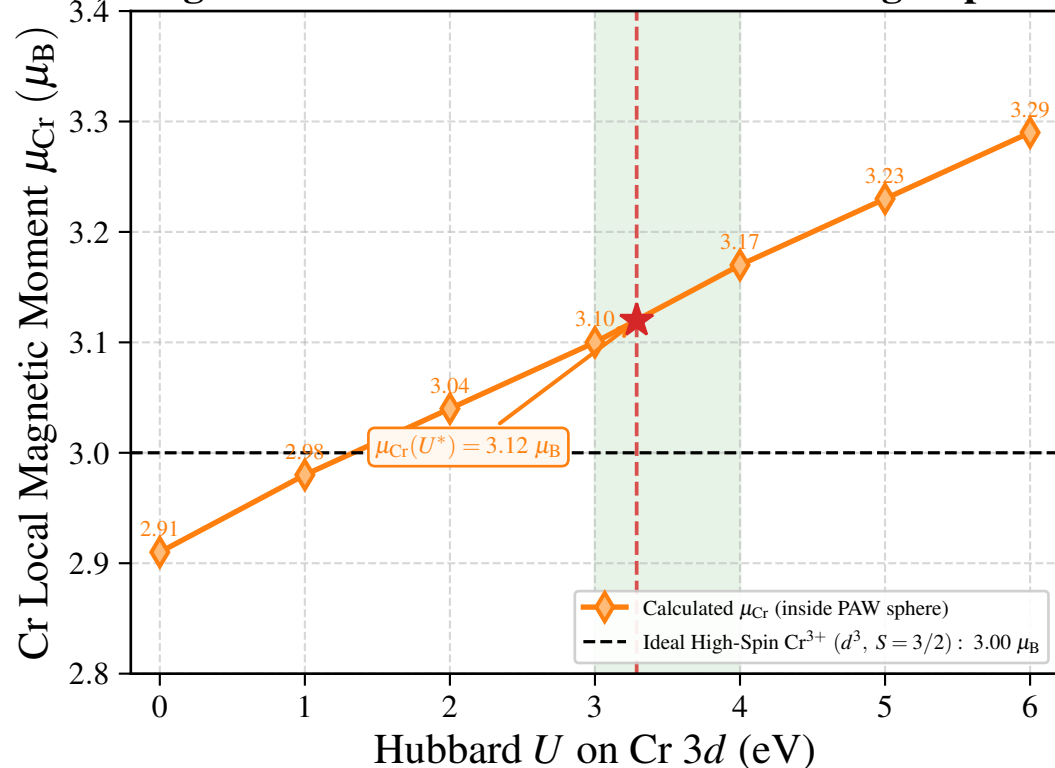
(a) Structural Constraint



(b) Electronic Constraint



(c) Magnetic Moment: Localization of 3d<sup>3</sup> High-Spin State



## Quantitative Synthesis & Recommendation for $2 \times 2 / 3 \times 3$ :

### 1. Exact Structural Intersection with Webster 2018:

- $U^* = 3.29$  eV yields  $a = 6.0560$  Å (0.00% error vs Webster 2018 & Luo 2020)
- $E_g(U^*) = 2.56$  eV (matches quasi-particle charge-transfer opening)
- $\mu_{Cr}(U^*) = 3.12 \mu_B$  (localized  $S = 3/2$  high-spin magnetic state)

### 2. Production Recommendation for Fixed- $U$ Calculations:

[Option A]  $U = 3.0$  eV (Lowest Structural Bias)

→  $a = 6.052$  Å (−0.06% vs Webster),  $E_g = 2.54$  eV,  $\mu_{Cr} = 3.10 \mu_B$

[Option B]  $U = 4.0$  eV (Standard Trihalide Benchmark)

→  $a = 6.065$  Å (+0.15% vs Webster),  $E_g = 2.59$  eV (peak CT gap)

→ Cococcioni linear-response consensus for Cr 3d in halides ( $\sim 3.8$  eV)

→ Matches pre-staged benchmark sites (H\_S1\_U4, H\_S2\_U4)

### 3. Recommended Scientific Strategy for Manuscript:

- Compute full parametric series  $U \in [0, 6]$  eV (already staged in crcl3-2x2)
- Present  $U = 3.0$  eV or  $U = 4.0$  eV as the primary baseline in the main text
- Report  $U = 0 \dots 6$  eV as sensitivity analysis proving HER activity robustness!